

PROJECT VARUNA-NET 2.0: The Unified Ground–Space–Edge AI Fusion Grid

A Multi-Scale Hydro-Informatics & Disaster Architecture Integrating Hillslope Landslide Factor of Safety ($FS < 1.0$), Catchment Hydrology, 1D Saint-Venant PI-GNN Stormwater Modeling, Dynamic Arabian Sea Tidal Lock, and Flood-Aware GraphHopper Routing

Target Competition / Track:	e-Yantra Innovation Challenge (eYIC) · Hydro-Informatics & Edge AI Track
Institutional Alignment:	MoES (Mission Mausam) · CWC India-WRIS · NDMA National Disaster Management
Sensing Triad:	Space (NISAR L+S, EOS-04, Sentinel-1) + Radar (IMD DWR) + Ground (CRNS Mesoscale)
Edge Compute & Telemetry:	Qualcomm Dragonwing QCS6490 / RB3 Gen 2 · 3GPP Rel-17 NTN Sat-IoT + CRSN TVWS
Hydrodynamic & AI Engines:	1D Saint-Venant PI-GNN (sub-4s) + Arabian Sea Tidal Lock + GraphHopper B2B Routing
Release Version & Date:	Integrated Master Version 2.1 · September 2026

Executive Summary & The Unified Vision: India's hydro-meteorological crisis does not exist in silos. During extreme monsoon convective storms, precipitation triggers severe **hillslope translational landslides** in upstream ghats and valley slopes at the exact same moment that surging **saturation-excess runoff** cascades downstream to submerge urban stormwater networks in under 60 minutes (as observed in Mumbai, Wayanad, Chennai, and Himalayan foothill corridors). Existing government platforms (CWC's ~360 river gauge network, IMD's ~47–50 radar grid) address these disasters independently. **Project VARUNA-NET 2.0** unifies these disciplines into a single continuous physics-and-AI pipeline. By deploying **Cosmic-Ray Neutron Sensing (CRNS)**, the system measures the single master physical state variable linking both hazards: antecedent vadose-zone soil saturation (θ). On steep hillslopes, CRNS drives edge-computed geotechnical **Factor of Safety ($FS < 1.0$)** slope failure alerts. In peri-urban catchments (such as Sanjay Gandhi National Park and the Mithi River headwaters), CRNS data is injected as a dynamic boundary condition into an ultra-fast (<4 sec) **Physics-Informed Graph Neural Network (PI-GNN)** that solves 1D Saint-Venant hydrodynamic equations across municipal sewer graphs. Crucially, the architecture integrates the **Arabian Sea Tidal Lock Mechanism** on coastal outfalls (>4.2m astronomical spring tides corking gravity drainage) and couples with a **Flood-Aware GraphHopper Routing Engine** that dynamically penalizes road weights $W_{ij} = W_{base} + P(d)$ for B2B fleet logistics (Swiggy, Uber) and emergency dispatches. Backed by **NASA-ISRO NISAR** satellite radar and **3GPP Release-17 NTN Satellite-IoT** failover, VARUNA-NET 2.0 provides an end-to-end national disaster intelligence grid.

1. The Multi-Hazard Gap Analysis & Architectural Unification

Previous disaster platforms suffered from an artificial divide: landslide monitoring operated purely on mountain slopes without considering downstream flood hydrographs, while urban flood nowcasting modeled city stormwater pipes without knowing upstream catchment antecedent moisture or ocean tidal locking. VARUNA-NET 2.0 closes this gap completely:

Capability Dimension	Landslide Geotech Framework	Mumbai PI-GNN Framework	Unified VARUNA-NET 2.0 Grid
Target Hazards	Hillslope landslides & natural river discharge	Urban pipe surcharge & coastal street ponding	Unified Cascade: Hillslope failure + River discharge + 1D Saint-Venant surcharge + Tidal backflow
CRNS Sensor Role	Computes pore-water pressure $u(\theta)$ and Mohr-Coulomb shear loss	Assumes fixed boundary inflow hydrographs	Dual Physics Engine: Evaluates slope FS and injects upstream saturation into Mithi River boundary $Q_{\text{boundary}}(\theta)$
Coastal Boundary	Not considered (inland mountainous terrain)	Semi-diurnal tides corking 186 outfalls (>4.2m lock)	Coupled Coastal Engine: Dynamic astronomical tide solver controlling flap gates & pump station capacity
Space & Radar Segment	Ground rain gauges & tiltmeters only	IMD Doppler Radar (DWR) ConvLSTM nowcast	Tri-SAR + Radar: NISAR (L+S) + Sentinel-1 + IMD Radar blended with in-situ CRNS ground truth
Routing & B2B Logistics	Static evacuation route warnings	GraphHopper flood routing API (P(d) penalties)	Dynamic Fleet Navigation: Sub-second GraphHopper edge weight update for Swiggy/Uber & emergency fleets
Communication Layer	Cognitive Radio (CRSN TV White Space in valleys)	Standard cellular 4G/5G backhaul	Triple Redundancy: 4G/5G primary + 3GPP Rel-17 NTN Sat-IoT + CRSN TV White Space in gorges

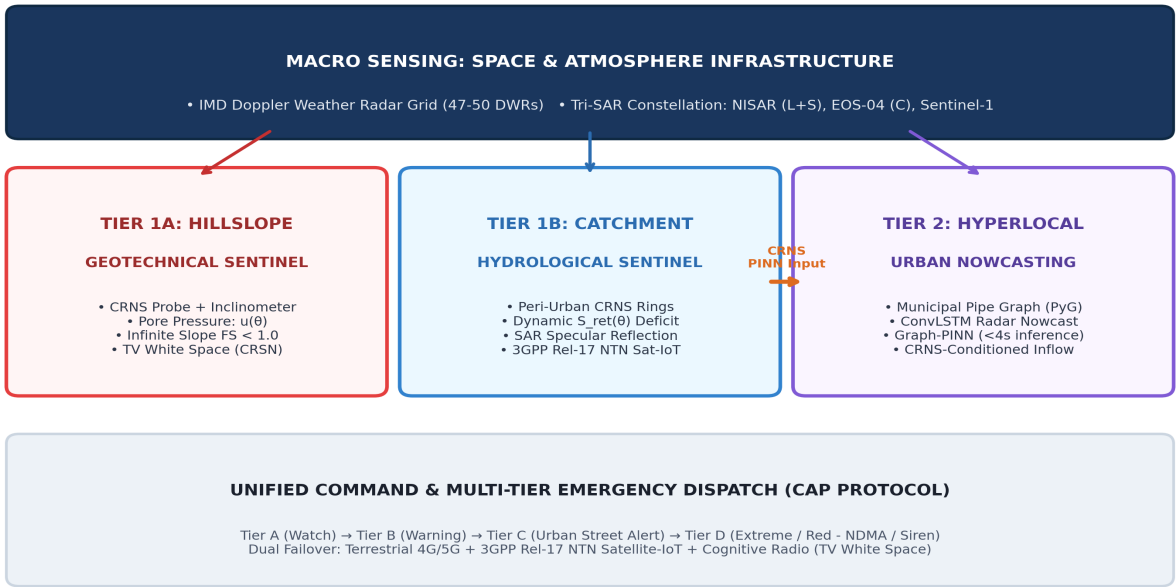


Figure 1: Project VARUNA-NET 2.0 System Architecture connecting Space Radar, Hillslope Sentinels, Peri-Urban Catchments, Urban PI-GNN Surrogates, Arabian Sea Tidal Gates, and GraphHopper B2B Fleet Navigation.

2. Technical Stack, Geotechnical Physics & Hydrodynamics

2.1 Tier 1A: Hillslope Geotechnical Sentinel (Landslides)

Deployed on steep, metastable slopes (e.g., Western Ghats ghats, Himalayan corridors, and Mumbai's urban hillocks in Ghatkopar/Malad). CRNS neutron counts are converted via the Desilets equation and evaluated dynamically against the **Infinite Slope Stability Criterion**:

$$FS(t) = [c' + (\gamma_{bulk}(\theta) \cdot z \cdot \cos^2\beta - u(\theta)) \cdot \tan\phi'] / [\gamma_{bulk}(\theta) \cdot z \cdot \sin\beta \cdot \cos\beta]$$

Where pore-water pressure $u(\theta) = m(\theta) \cdot \gamma_w \cdot z \cdot \cos^2\beta$ activates as CRNS saturation $S_r = \theta / \theta_{sat}$ exceeds 0.70. Dynamic Rainfall Intensity-Duration (I-D) thresholds are updated: $I_{crit}(D, \theta) = \alpha \cdot [1 - (\theta / \theta_{sat})]^\gamma \cdot D^{(-\beta_{slope})}$, suppressing false alarms on dry soils.

2.2 Tier 1B: Macro-Catchment Hydrology & Mithi River Boundary Coupling

Deployed in peri-urban catchments (e.g., Sanjay Gandhi National Park, Vihar and Tulsi lakes). The potential retention capacity S_{ret} is continuously updated by the CRNS moisture deficit: $S_{ret}(t) = S_{max} \cdot [1 - (\theta_{CRNS}(t) / \theta_{sat})]$. Direct surface runoff Q_{excess} is calculated and injected as a dynamic boundary inflow hydrograph into the Mithi River inlet nodes of Mumbai's urban drainage network:

$$Q_{boundary}(t, \theta) = Q_{base} + [(P_{eff}(t)^2 / (P_{eff}(t) + S_{ret}(\theta))) \cdot A_{catchment} \cdot (\theta / \theta_{sat})^2] / 3.6$$

2.3 Tier 2: 1D Saint-Venant Physics-Informed Graph Neural Network (PI-GNN)

Municipal stormwater conduits are parsed as a directed graph $G = (V, E)$. Instead of waiting 60–90 minutes for numerical SWMM/HEC-RAS solvers to converge, an edge-based **Physics-Informed Graph Neural Network (PI-GNN)** solves 1D sewer hydrodynamics in **sub-4-second inference** by embedding the full 1D Saint-Venant Partial Differential Equations directly into its loss function:

$$\text{Continuity (Mass Conservation): } \partial A / \partial t + \partial Q / \partial x - q_L = 0$$

$$\text{Momentum Conservation: } \partial Q / \partial t + \partial (Q^2 / A) / \partial x + g \cdot A \cdot (\partial h / \partial x) - g \cdot A \cdot (S_0 - S_f) = 0$$

Where Manning friction slope $S_f = n^2 \cdot |Q| \cdot Q / (A^2 \cdot R^{4/3})$. The total loss function optimized during training is:

$$L_{total} = \lambda_{data} \cdot L_{data} + \lambda_{mass} \cdot L_{mass} + \lambda_{mom} \cdot L_{mom} + \lambda_{CRNS} \cdot ||Q_{inlet} - Q_{boundary}(\theta_{CRNS})||^2$$

2.4 Dynamic Arabian Sea Tidal Lock Mechanism & 186 Outfall Sluice Flap Gates

Mumbai's drainage grid relies on 186 outfalls discharging directly into the Arabian Sea, Mahim Creek, and Thane Creek. During astronomical spring tides, the sea level swings up to +4.8m CD. Whenever the tide height exceeds the outfall invert level ($h_{tide} \geq 4.2m$), heavy steel flap gates slam shut to prevent sea water from backflowing into the city. This completely corks gravity drainage ($Q_{gravity} = 0 \text{ m}^3/s$), meaning incoming rainwater and Mithi River discharge must be evacuated entirely by mechanical BMC dewatering pump stations (e.g., Haji Ali, Love Grove, Britannia, Irla, Gazdarbandh). VARUNA-NET 2.0 explicitly incorporates the dynamic downstream coastal boundary condition:

$$h_{outfall}(t) = \max(h_{pipe}(t), h_{tide}(t)), \quad Q_{outfall}(t) = Q_{pump} + (1 - \text{Gate_Locked}(t)) \cdot Q_{gravity}(h_{pipe} - h_{tide})$$

2.5 Flood-Aware GraphHopper Routing Engine with Dynamic Road Edge Penalties

Predicted 2D overland water depths $d(x, y, t)$ are mapped in real time onto the urban road network graph $G_{road} = (V, E)$. Each road edge weight W_{ij} is dynamically recomputed to provide actionable rerouting for quick-commerce (Swiggy, Zomato, Zepto), ride-hailing (Uber, Ola), and municipal emergency fleets (ambulances, BMC pump trucks):

$$W_{ij}(d) = W_{base} + P(d), \quad \text{where } P(d) = 0 \text{ (} d < 5cm \text{)} \mid W_{base} \cdot [1 + 0.40(d - 5)] \text{ (} 5 \leq d < 15cm \text{)} \mid \infty \text{ (} d \geq 15cm \text{ - Severed)}$$

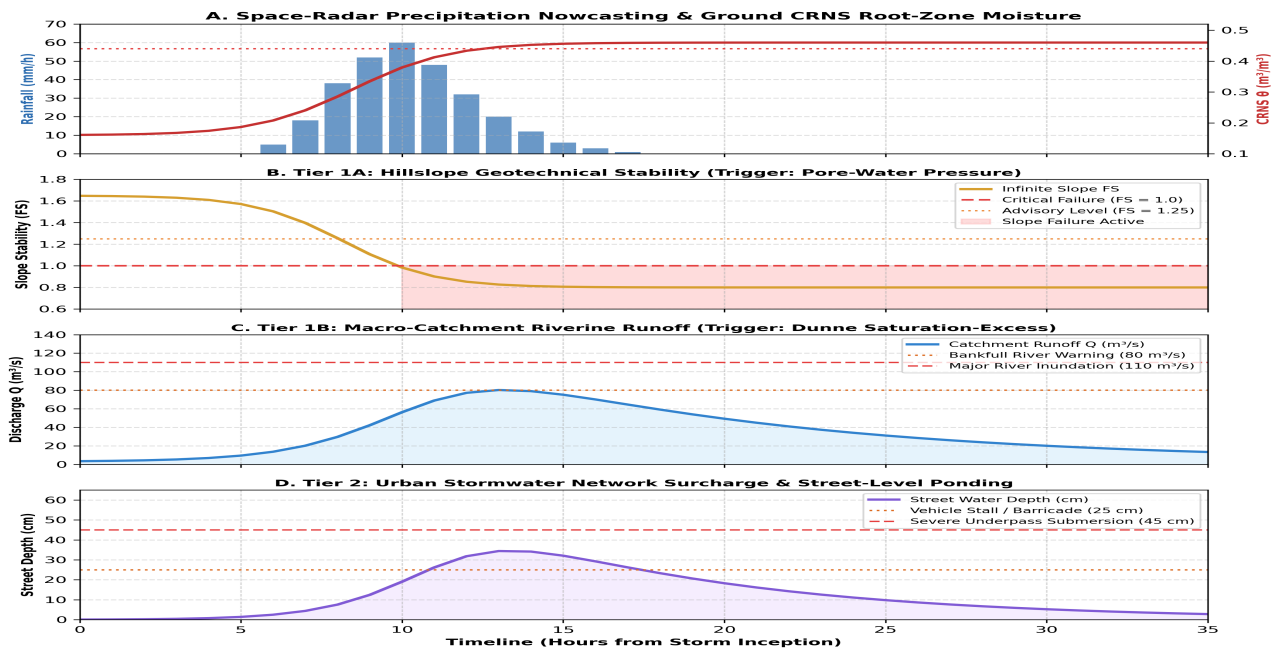


Figure 2: 36-Hour Multi-Hazard Simulation: Convective storm driving simultaneous hillslope FS collapse (Landslide), catchment discharge surge (Riverine Flood), and street-level pipe surcharge (Urban Pluvial Inundation).

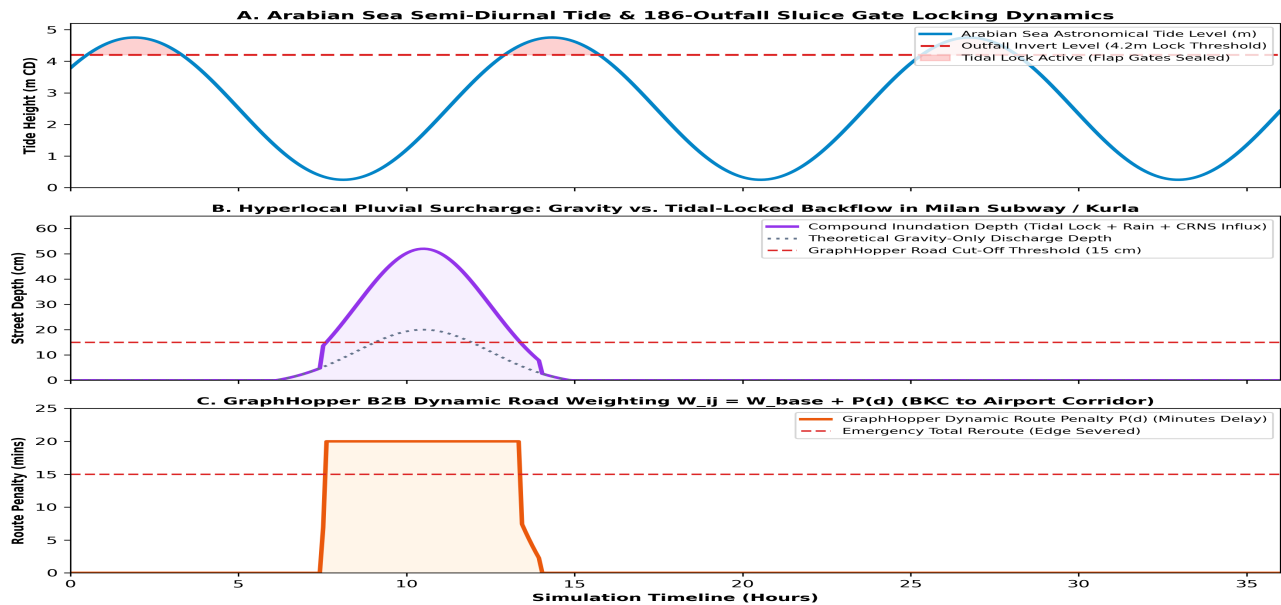


Figure 3: Mumbai Coastal Coupling Analysis: (A) Semi-diurnal Arabian Sea Tide & 4.2m Outfall Sluice Gate Locking; (B) Severe Pluvial Surge under Compound Rain + Tidal Lock; (C) GraphHopper Road Edge Routing Penalties $P(d)$.

3. Multi-Tier Alerting Logic & CAP Emergency Dispatch

VARUNA-NET 2.0 synthesizes multi-hazard triggers into a unified 4-tier alerting protocol aligned with the National Disaster Management Authority (NDMA) and State Disaster Management Authorities (SDMAs):

Alert Tier	Trigger Conditions (Multi-Hazard Logic)	Target Stakeholders & Actionable Protocols
Tier A WATCH	• CRNS neutron-drop anomaly detected ($\theta/\theta_{sat} > 0.70$). • No radar cloudburst or SAR confirmation yet. • Slope FS between 1.30 and 1.50. • Astronomical tide < 3.5m CD.	• SDMA & municipal situational dashboard notification. • Pre-position mobile dewatering pumps in low-lying wards. • Routine 15-minute telemetry polling.
Tier B WARNING	• CRNS saturation > 0.80 + SAR flood polygon confirmation. • Hillslope FS drops into Advisory zone ($1.10 < FS \leq 1.30$). • Radar nowcast predicts >30 mm/h incoming rain. • Tide rising above 3.8m CD.	• District Administration, Irrigation Dept & Traffic Police. • Issue heavy traffic advisories along flood-prone arterial roads. • Pre-activate BMC dewatering stations (Haji Ali, Love Grove).
Tier C NOWCAST ALERT	• PI-GNN predicts street inundation > 15 cm within 60–120 min. • Spring high tide > 4.2m triggers Outfall Tidal Lock. • Hillslope FS drops to 1.02–1.10 (Marginal Stability).	• Municipal Control Room & automated traffic diversion. • GraphHopper API activates automated B2B route detours. • Deploy physical barricades at underpasses (e.g., Milan Subway).
Tier D EXTREME / RED	• Landslide Failure Imminent: Hillslope FS ≤ 1.00. • Catastrophic Inundation: Street depth > 45 cm under complete Tidal Lock. • CWC river gauge crosses Highest Flood Level (HFL).	• NDRF / SDRF Full Mobilization per Red Bulletin protocol. • Automated siren sounding in vulnerable hillside slums & floodplains. • 3GPP Rel-17 NTN Satellite Broadcast if cellular towers fail.

4. Triple-Redundant Telemetry Backbone

Communication breakdown is the single most common failure point during extreme cloudbursts. VARUNA-NET 2.0 integrates a **Triple-Redundant Telemetry Architecture**:

1. **Primary (Terrestrial 4G/5G):** Transmits high-frequency 1-minute telemetry and edge embeddings via MQTT over TLS.
2. **Secondary (3GPP Release-17 NTN Satellite-IoT):** When cell towers lose power or backhaul, the Qualcomm Dragonwing gateway automatically switches to direct-to-GEO satellite modems, beaming 1 kB binary CBOR alert packets.
3. **Tertiary (Cognitive Radio / TV White Space):** In deep Himalayan or Western Ghats ravines where line-of-sight to satellites is obstructed by steep canyon walls, CRSN nodes opportunistically hop across VHF/UHF TV White Space (470–698 MHz) to relay packets to the nearest ridge-top gateway.

5. Hardware Specification, Budget & Dual-Path Strategy

Component Layer	Prototyping Path (eYIC Budget)	Production Field Deployment	Estimated Cost Envelope
Neutron Sensing (CRNS)	ESP32-based Monte Carlo physics simulator generating calibrated pulse counts	Commercial Helium-3 / Boron-lined proportional counter tube	Simulator: ~₹1,500 Field Tube: ₹3.5L – ₹6.0L
Environmental & Ground	BME280 (P, T, RH) + MPU6050 tiltmeter	Industrial ultrasonic weather hub + Geotechnical vibrating wire piezometer	Proto: ~₹1,200 Field: ~₹45,000
Edge AI Compute	Raspberry Pi 5 (8GB) running INT8 ONNX Runtime	Qualcomm Dragonwing QCS6490 / RB3 Gen 2 (12 TOPS Hexagon NPU)	Proto: ~₹8,500 Field: ~₹35,000 (volume)
Satellite-IoT Modem	Waveshare SIM7080G / NTN DevKit	3GPP Rel-17 NTN certified satellite transceiver module	Proto: ~₹4,500 Field: ~₹15,000
Power Autonomy	12V 20Ah Lead-Acid + 50W solar panel	100W Monocrystalline + 12V 100Ah LiFePO4 battery (14 days autonomy)	Proto: ~₹6,000 Field: ~₹28,000

6. Project Implementation Roadmap (8-Month eYIC Plan)

Phase & Timeline	Key Milestone Objectives	Verifiable Performance Target
Phase 1 (Months 1–2) Hardware & Graph Baseline	- Assemble ESP32 simulator + BME280 + Raspberry Pi 5 / RB3 Gen 2. - Ingest municipal storm drain shapefiles and construct PyTorch Geometric graph. - Connect to IMD Doppler Radar (DWR) and ISRO Bhuvan CartoDEM feeds.	Drainage graph topology verified; radar reflectivity raster ingestion latency < 30 s.
Phase 2 (Months 3–4) AI Surrogates & Slope Physics	- Train ConvLSTM precipitation nowcaster (0–3h forward horizon). - Code and calibrate Infinite Slope Stability (FS) & Desilets CRNS modules. - Train 1D Saint-Venant PI-GNN graph surrogate on historical SWMM hydrodynamic runs.	PI-GNN inference < 4 s per ward (99% speedup over SWMM); Landslide FS accuracy verified.
Phase 3 (Months 5–6) Tidal Lock & NTN Failover	- Integrate dynamic Arabian Sea tidal lock solver & outfall flap gate logic. - Couple Tier 1B CRNS saturation as dynamic boundary condition in PI-GNN loss. - Implement GraphHopper B2B routing engine with dynamic road edge penalties P(d). - Execute benchtop power-cut simulation to validate 3GPP Rel-17 NTN satellite failover.	100% packet delivery to cloud within 60 s under complete cellular tower blackout; dynamic reroute latency < 500 ms.
Phase 4 (Months 7–8) Field Trial, Dashboard & Pitch	- Build real-time command dashboard showing multi-hazard alerts, tidal lock status & fleet routes. - Conduct controlled catchment saturation test (irrigation-induced wetting). - Replay historical cloudburst events (Mumbai 2005/2024, Wayanad 2024, Chennai 2015/2023).	Overall false-positive reduction > 65%; pitch demonstration ready for eYIC evaluation.

7. Novelty Statement for eYIC / MoES Evaluation

What Makes VARUNA-NET 2.0 Superior:

- Multi-Hazard Physics Unification:** Unifies hillslope landslides and urban stormwater surcharges under a single physical master variable (CRNS antecedent vadose-zone moisture θ).
- 1D Saint-Venant PI-GNN with Upstream CRNS Coupling:** Replaces slow 60-minute hydrodynamic models with sub-4-second physics-informed neural surrogates explicitly driven by upstream catchment saturation.
- Dynamic Coastal Tidal Lock Mechanics:** Directly models the real-life failure mode of Mumbai's 186 outfalls where >4.2m spring tides seal flap gates, triggering severe pluvial backflow.

4. **Actionable B2B GraphHopper Routing:** Bridges the last-mile gap from alert issuance to commercial impact, providing dynamic road edge penalties $W_{ij} = W_{base} + P(d)$ for quick-commerce, ride-hailing, and emergency ambulances.
5. **NISAR + NTN Space Integration:** Fully grounds remote sensing in the operational 2025/2026 NASA-ISRO NISAR mission (L+S band) and 3GPP Release-17 NTN satellite-IoT standards.